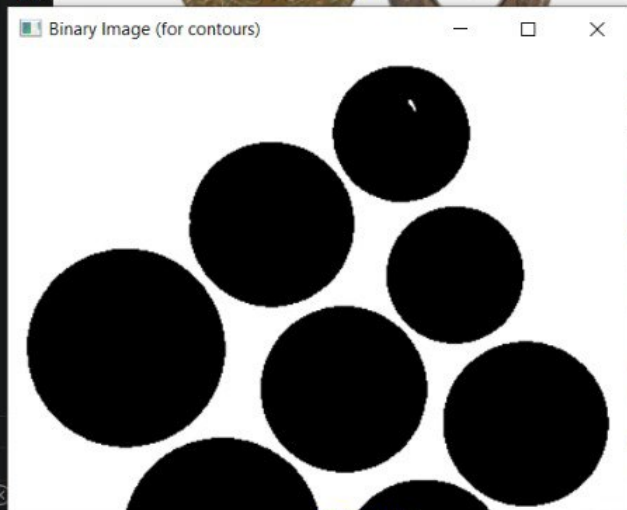
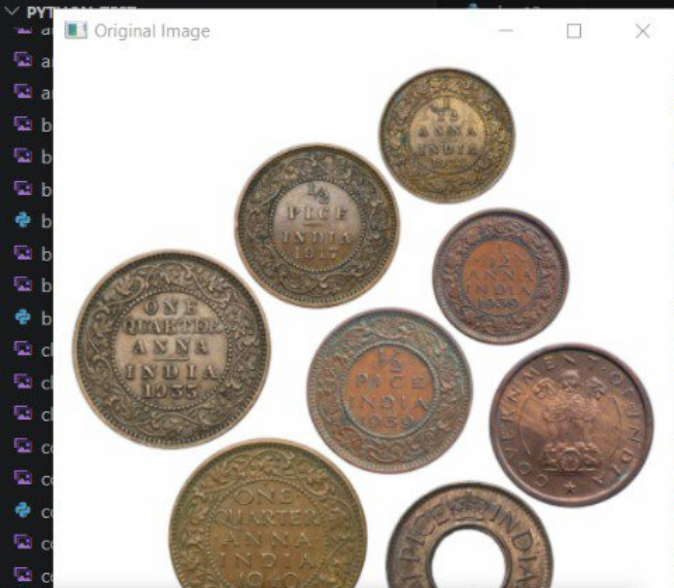




```
reference, "CONTOUR REFERENCE GUIDE", (ref_width//2 - 230, 100, cv2.FONT_HERSHEY_SIMPLEX, 0.9, (255, 255, 255), 2)

# Contours visualization
image = np.zeros((150, 150), dtype="uint8")
contour_example, (75, 75), 60, 255, -1)
_, _ = cv2.findContours(contour_example, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)[1:]
_, _ = cv2.findContours(contour_example, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)

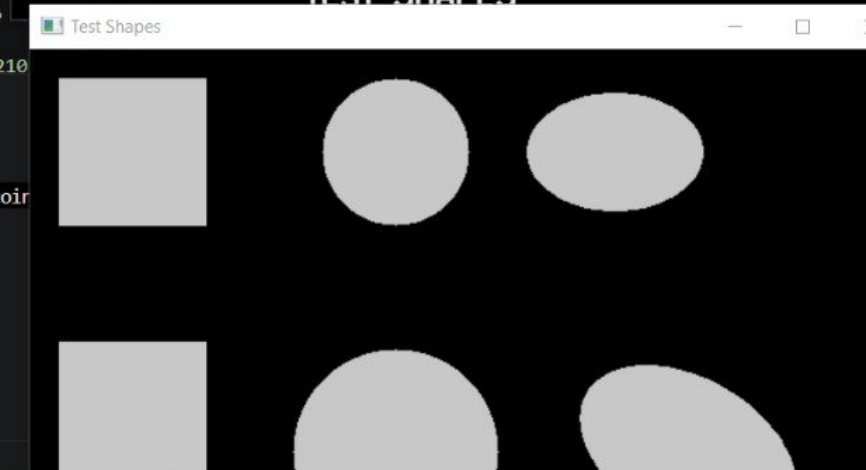
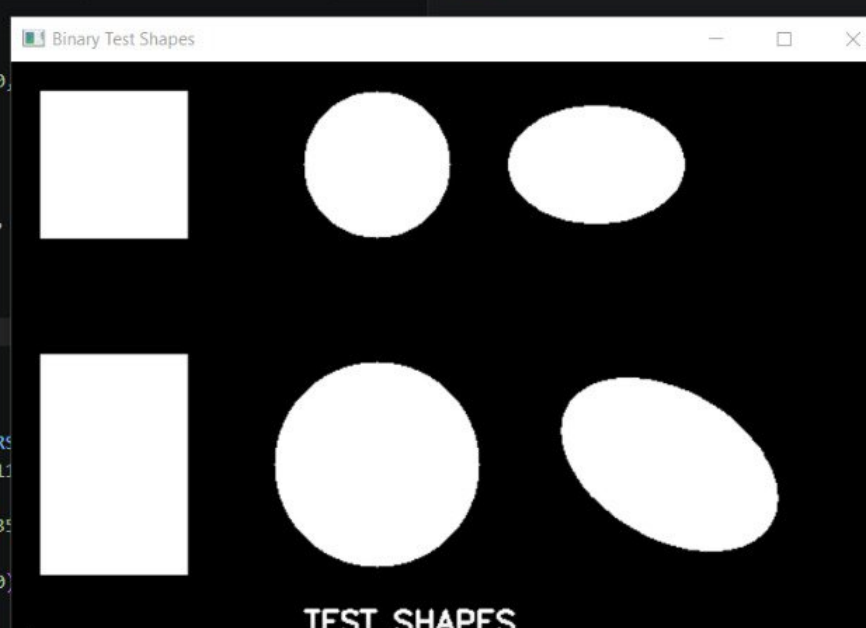
reference, "CONTOUR PROPERTIES:", (200, 80), cv2.FONT_HERSHEY_SIMPLEX, 0.5, (200, 200, 200), 1)
reference, "cv2.contourArea(c) - Area of contour", (200, 110), cv2.FONT_HERSHEY_SIMPLEX, 0.5, (200, 200, 200), 1)
reference, "cv2.arcLength(c, True) - Perimeter", (200, 135), cv2.FONT_HERSHEY_SIMPLEX, 0.5, (200, 200, 200), 1)
reference, "cv2.boundingRect(c) - Bounding box", (200, 160), cv2.FONT_HERSHEY_SIMPLEX, 0.5, (200, 200, 200), 1)
reference, "cv2.minEnclosingCircle() - Enclosing circle", (200, 185), cv2.FONT_HERSHEY_SIMPLEX, 0.5, (200, 200, 200), 1)
reference, "cv2.moments(c) - Center calculation", (200, 210), cv2.FONT_HERSHEY_SIMPLEX, 0.5, (200, 200, 200), 1)

DEBUG CONSOLE TERMINAL PORTS

python day13_contours.py --image coin

ing image and creating binary image...
7 pixels

age with known shapes...
```



Python_Test

Sign In

EXPLORER

PYTHON_TEST

day11_thresholding.py

day12_gradients_edges.py

day13_contours.py

dissolve.mp4

drawin_basics.py

drawing.py

edge_directions.png

edge_preserving_comparison.png

equalization_comparison.png

equalize.py

equalized_image.png

ex_1_day_3.py

ex_2_day_3.py

ex_3_day_3.py

ex_4_day_3.py

exercise_1_contour_analysis.py

exercise_1_custom_pipeline.py

exercise_1_directional_edges.py

exercise_1_house.py

exercise_1.py

exercise_2_auto_canny.py

exercise_2_comparison.py

exercise_2_target.py

exercise_2.py

exercise_3_chessboard.py

exercise_3_document_edges.py

exercise_3.py

exercise_4_annotation.py

exercise_4_sharpness_analysis.py

exercise_5_clock.py

exercise_5_object_counting.py

exercise_5.py

outline

timeline

exercise_1_contour_analysis.py

4 def analyze_contours(image_path):

47 print("-" * 50)

48 print(f"Total contours: {len(contours)}")

49

50 if len(areas) > 0:

51 print(f"Total area: {sum(areas):.0f} pixels")

52 print(f"Average area: {np.mean(areas):.0f}")

53 print(f"Largest: {max(areas):.0f}")

54 print(f"Smallest: {min(areas):.0f}")

55 else:

56 print("No contours found.")

57

58 return contours, areas, perimeters

59

60

61 # Run analysis

62 analyze_contours("test_image.png")

63

64 cv2.destroyAllWindows()

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Contour 1: Area=9082, Perimeter=409.8

Contour 2: Area=9082, Perimeter=409.8

Contour 3: Area=9660, Perimeter=471.7

Contour 4: Area=9660, Perimeter=471.7

Total contours: 4

Total area: 37482 pixels

Average area: 9370

Largest: 9660

Smallest: 9082

PS C:\Users\jebtron\jason\Desktop\Python_Test>

powerShell

powerShell

powerShell

powerShell

powerShell

powerShell

powerShell

powerShell

powerShell

powerShell

CHAT

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

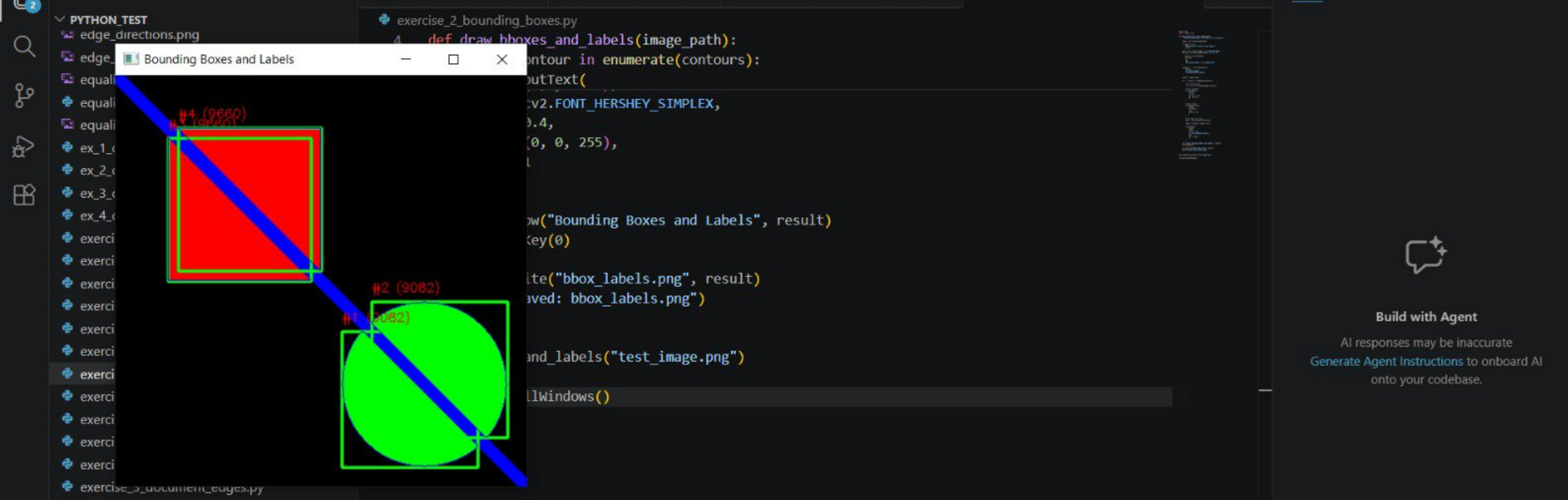
exercise_1_contour_analysis.py

Describe what to build

Local Default Approvals

Ln 64, Col 24 Spaces: 4 UTF-8 CRLF Python

33°C Light rain 17:27 18-06-2026



PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Contour 1: Area=9082, Perimeter=409.8
Contour 2: Area=9082, Perimeter=409.8
Contour 3: Area=9660, Perimeter=471.7
Contour 4: Area=9660, Perimeter=471.7
-----
Total contours: 4
Total area: 37482 pixels
Average area: 9370
Largest: 9660
Smallest: 9082
PS C:\Users\jebtron\jason\Desktop\Python_Test> python exercise_2_bounding_boxes.py
```

Ln 78, Col 24 Spaces: 4 UTF-8 CRLF {} Python

Python_Test

File Edit Selection View Go Run Terminal Help

EXPLORER

PYTHON_TEST

equalized_image.png

ex_1_day_3.py

ex_2_day_3.py

ex_3_day_3.py

ex_4_day_3.py

exercise_1_contour_analysis.py

exercise_1_custom_pipeline.py

exercise_1_directional_edges.py

exercise_1_house.py

exercise_1.py

exercise_2_auto_canny.py

exercise_2_bounding_boxes.py

exercise_2_comparison.py

exercise_2_target.py

exercise_2.py

exercise_3_chessboard.py

exercise_3_crop_objects.py

exercise_3_document_edges.py

exercise_3.py

exercise_4_annotation.py

exercise_4_sharpness_analysis.py

exercise_5_clock.py

exercise_5_object_counting.py

exercise1_brightness_contrast.py

exercise1_channel_art.py

exercise1_histogram_analyzer.py

exercise1_interactive_blur.py

exercise1_threshold_optimizer.py

exercise1_thumbnail_gallery.py

exercise2_color_picker.py

exercise2_document_hinarization.py

OUTLINE

TIMELINE

day13_contours.py

exercise_1_contour_analysis.py

exercise_2_bounding_boxes.py

exercise_3_crop_objects.py

exercise_3_crop_objects.py

```
5 def crop_and_save_objects(image_path, output_dir="cropped_objects"):
36     for i, contour in enumerate(contours):
52         # Save
53         filename = os.path.join(
54             output_dir,
55             f"object_{i+1}.png"
56         )
57
58         cv2.imwrite(filename, obj)
59
60         print(f"Saved: {filename}")
61
62     print(f"\nAll objects saved to '{output_dir}'")
63
64
65 # Run cropping
66 crop_and_save_objects("test_image.png")
67
68 cv2.destroyAllWindows()
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS C:\Users\jeb_\jason\Desktop\Python_Test> python exercise_2_bounding_boxes.py

Saved: bbox_labels.png

PS C:\Users\jeb_\jason\Desktop\Python_Test> python exercise_3_crop_objects.py

PS C:\Users\jeb_\jason\Desktop\Python_Test> python exercise_3_crop_objects.py

Cropping 4 objects...

Saved: cropped_objects\object_1.png

Saved: cropped_objects\object_2.png

Saved: cropped_objects\object_3.png

Saved: cropped_objects\object_4.png

All objects saved to 'cropped_objects'

PS C:\Users\jeb_\jason\Desktop\Python_Test>

+ powershell

+ powershell

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+ powershell

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+ powershell

+ powershell

+ exercise_3_crop_objects.py

Describe what to build

+ | | Auto |

Local | Default Approvals

Sign In

CHAT

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

Ln 65, Col 15

Spaces: 4

UTF-8

CRLF

{ } Python

17:32

18-06-2026

Python_Test

File Edit Selection View Go Run Terminal Help

EXPLORER

PYTHON_TEST

exercise_1_contour_analysis.py

exercise_1_custom_pipeline.py

exercise_1_

exercise_1_

exercise_1_

exercise_2_

exercise_2_

exercise_2_

exercise_2_

exercise_2_

exercise_2_

exercise_3_

exercise_3_

exercise_3_

exercise_3_

exercise_4_

exercise_4_

exercise_4_

exercise_5_

exercise_5_

exercis1_

exercis1_

exercis1_

exercise1_interactive_blur.py

exercise1_threshold_optimizer.py

exercise1_thumbnail_gallery.py

exercise2_color_picker.py

exercise2_document_binarization.py

exercise2_equalization_grid.py

exercise2_image_dissolve.py

exercise2_noise_reduction.py

exercise2_panorama_prep.py

OUTLINE

TIMELINE

contour_analysis.py

exercise_2_bounding_boxes.py

exercise_3_crop_objects.py

exercise_4_shape_classifier.py

exercise_4_shape_classifier.py

4 def classify_shapes(image_path):

105 print("-" * 40)

count in shape_counts.items():

{shape}: {count}")

Classified Shapes", result)

0)

"classified_shapes.png", result)

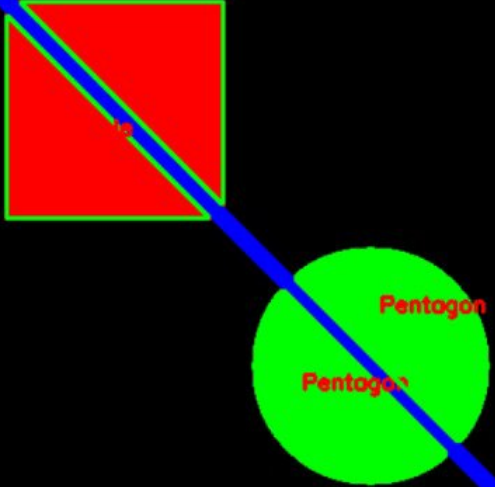
: classified_shapes.png")

sification

"test_image.png")

ndows()

Classified Shapes



PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Saved: cropped_objects\object_1.png

Saved: cropped_objects\object_2.png

Saved: cropped_objects\object_3.png

Saved: cropped_objects\object_4.png

All objects saved to 'cropped_objects'

PS C:\Users\jebroon\jason\Desktop\Python_Test> python exercise_4_shape_classifier.py

Shape Classification Results:

Pentagon: 2

Triangle: 2

+ powershell

+ powershell

+ powershell

+ powershell

+ powershell

+ powershell

+ powershell

+ powershell

+ powershell

+ python

+ exercise_4_shape_classifier.py

Describe what to build

+ Auto

Local

Default Approvals

Type here to search

33°C Light rain

17:34

18-06-2026

Python_Test

File Edit Selection View Go Run Terminal Help

EXPLORER

PYTHON_TEST

exercise_1_house.py

exercise_1.py

exercise_2_auto_canny.py

exercise_2_bou

exercise_2_com

exercise_2_targ

exercise_2.py

exercise_3_che

exercise_3_cro

exercise_3_doc

exercise_3.py

exercise_4_ann

exercise_4_shap

exercise_4_shap

exercise_5_cloc

exercise_5_doc

exercise_5_obje

exercise1_bright

exercise1_chan

exercise1_histo

exercise1_inter

exercise1_thres

exercise1_thum

exercise2_color_picker.py

exercise2_document_binarization.py

exercise2_equalization_grid.py

exercise2_image_dissolve.py

exercise2_noise_reduction.py

exercise2_panorama_prep.py

exercise3_3d_color_histogram.py

exercise3_color_isolation.py

OUTLINE

TIMELINE

bounding_boxes.py

exercise_3_crop_objects.py

exercise_4_shape_classifier.py

exercise_5_document_scanner.py

exercise_5_document_scanner.py

21

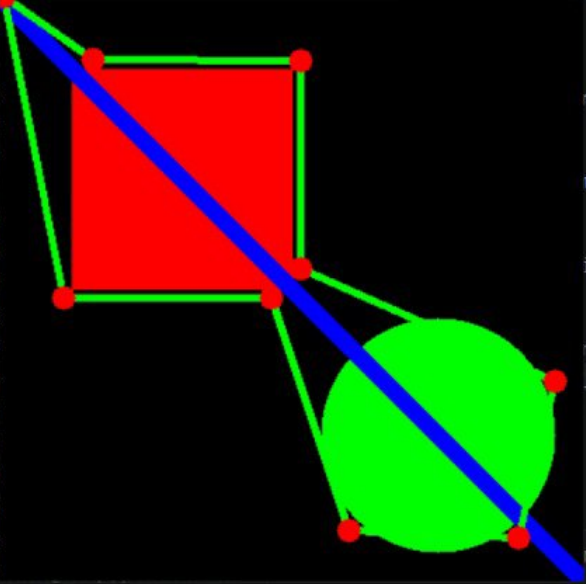
def scan_document(image_path):

72

if len(approx) == 4:

103

Document Detected



document.png",

ned document saved as scanned_document.png")

ment is not a quadrilateral. Cannot perform scan.")

indows()

ner

image.png")

TERMINAL

PS C:\Users\jeb_\Desktop\Python_Test> python exercise_4_shape_classifier.py

Shape Classification Results:

Pentagon: 2

Triangle: 2

Saved: classified_shapes.png

PS C:\Users\jeb_\Desktop\Python_Test> python exercise_5_document_scanner.py

Document corners found: 9

Document is not a quadrilateral. Cannot perform scan.

Local

Default Approvals

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

python

Describe what to build

Auto

Ln 119, Col 32

Spaces: 4

UTF-8

CRLF

{ } Python

17:35

18-06-2026

Type here to search

33°C Light rain

ENG